

Sai Teja Bandaru

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PROFESSIONAL SUMMARY

Generative AI Engineer with **5+ years** of experience designing and deploying production-grade LLM systems, agentic AI pipelines, and RAG architectures across enterprise environments. Deep expertise in **LangChain, LangGraph, OpenAI APIs, fine-tuning (LoRA/PEFT), and multi-agent orchestration**, with end-to-end delivery on Microsoft Azure. M.S. in Computer Science, University of Texas at Arlington. Industry background spanning healthcare, insurance, banking, and technology.

TECHNICAL SKILLS

Languages:	Python (Expert), SQL, Bash, JavaScript (basic)
GenAI & LLMs:	OpenAI API (GPT-3.5/4o), LLaMA-2/3, Mistral, Falcon, LangChain, LangGraph, OpenAI Agents SDK, LlamaIndex
Agentic AI:	Multi-agent workflows, tool-using LLMs, stateful memory, human-in-the-loop validation, dynamic tool routing
RAG & Search:	FAISS, Pinecone, Weaviate, Chroma — semantic search, embedding models, BM25, DPR, SBERT
Model Tuning:	LoRA, PEFT, Prompt Engineering (few-shot, zero-shot, chain-of-thought, reflection loops), Transfer Learning
ML Frameworks:	PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, LightGBM, Hugging Face Transformers
MLOps:	MLflow, Weights & Biases, DVC, CI/CD, GitHub Actions, Great Expectations, Prometheus, Grafana
Cloud:	Azure ML, Azure OpenAI, Azure Functions, Azure DevOps AWS S3 Google Cloud Storage
Deployment:	FastAPI, Flask, Docker, Kubernetes, ONNX, TorchScript
Data & NLP:	Pandas, NumPy, PySpark, Databricks, Apache Spark, spaCy, NLTK, OpenCV
Databases:	PostgreSQL, MongoDB, SQLite, Redis
Interpretability:	SHAP, LIME, Fairlearn

PROFESSIONAL EXPERIENCE

Gen AI Engineer

Client: Carda Health | Jun 2023 – Present

- Architected multi-agent AI systems using **LangChain, LangGraph, and OpenAI Agents SDK**, enabling autonomous decision-making pipelines that reduced manual review effort by ~40%.
- Designed and deployed production **RAG pipelines** with Pinecone, FAISS, and Weaviate, integrating source-citing agents to improve factual accuracy and cut hallucinations by 35%.
- Fine-tuned open-weight LLMs (Mistral, LLaMA, Falcon) using **LoRA and PEFT**, achieving up to 28% domain-specific accuracy gains on insurance and healthcare benchmarks.
- Engineered human-in-the-loop validation layers and response filters to ensure regulatory compliance across GenAI pipelines serving banking and healthcare clients.
- Delivered enterprise GenAI solutions on **Azure** (Azure OpenAI, Azure ML, Azure Functions), handling thousands of daily inference requests at p99 latency under 800ms.
- Applied advanced **prompt engineering** techniques — few-shot, chain-of-thought, and reflection loops — boosting LLM task consistency by 30% on structured output use cases.
- Implemented context window optimization and prompt chaining strategies for robust LLM orchestration across multi-turn conversational agents.

AI / ML Engineer

Client: Root Insurance | Jul 2022 – May 2023

- Built end-to-end ML pipelines for risk scoring, telematics underwriting, and sentiment analysis using **Python, XGBoost, scikit-learn, and PySpark** across insurance and healthcare domains.
- Engineered **RAG systems** with LangChain, FAISS, and Pinecone; integrated LLM function-calling via OpenAI Agents SDK and LangGraph for dynamic tool routing in production chatbots.
- Fine-tuned LLaMA, Falcon, and Mistral using **LoRA/PEFT**; implemented model routing strategies that reduced inference costs by 45% for latency-sensitive workloads.
- Deployed deep learning models (PyTorch, TensorFlow, Keras) for NER, text summarization, and tabular prediction, improving throughput 3x via ONNX export and batch inference.

- Implemented full **MLOps lifecycle** (MLflow, Great Expectations, Prometheus/Grafana, GitHub Actions CI/CD), cutting model deployment time from 2 weeks to 2 days.
- Containerized and orchestrated AI services with **Docker and Kubernetes**, delivering scalable microservices across banking and tech verticals.

Machine Learning Engineer

TCS — India | Jan 2020 – Dec 2021

- Developed ML models for risk scoring, churn prediction, and fraud detection using **Python, scikit-learn, and XGBoost**, improving fraud recall by 22% for insurance and banking clients.
- Built deep learning models (TensorFlow, Keras) for image classification and **OCR** in insurance document processing, reducing manual extraction time by 60%.
- Designed **NLP pipelines** for text classification, NER, and sentiment analysis using spaCy, NLTK, and custom embeddings, deployed on Azure ML via Azure Functions for real-time inference.
- Pioneered early semantic search and QA systems using TF-IDF, BM25, SBERT, and DPR for knowledge retrieval across healthcare and fintech products.
- Applied **SHAP, LIME, and Fairlearn** for model interpretability and bias audits to meet responsible AI standards for regulated health and insurance clients.
- Managed MLOps infrastructure on **Azure DevOps** with CI/CD pipelines, Docker/Kubernetes orchestration, and Databricks for distributed feature engineering.

EDUCATION

Master of Science in Computer Science

Jan 2022 – Dec 2023

University of Texas at Arlington — Texas, USA